NEBULA HUB

Project Presentation Speech

A page-wise speaking guide for the complete enterprise operations platform

Presenter 1	Presenter 2
HRMS, Recruitment, and AI features across all modules	Expenses and Inventory Management

Suggested duration: 18 to 25 minutes

Use this document as a natural speaking script. Do not read every sentence mechanically; explain the highlighted workflow while demonstrating the application.

Page 1 - Opening and Project Vision

Speaker: You

Speech: Good morning everyone. We are presenting Nebula HUB, an integrated enterprise operations platform designed to bring the major day-to-day activities of an organization into one connected system. Instead of maintaining separate tools for customer management, human resources, recruitment, expenses, and inventory, Nebula gives the organization a common platform with shared authentication, role-based permissions, tenant isolation, analytics, and AI assistance.

Our project is not only a collection of CRUD screens. The important idea is integration. A candidate hired through Recruitment can become an HRMS employee. Inventory equipment can be issued during employee onboarding. Expense and procurement workflows can be governed by budgets and approvals. This makes Nebula closer to a practical ERP hub than an isolated module demonstration.

The system uses a React and TypeScript frontend, a Node.js and Express backend, MongoDB for persistence, and a multi-tenant architecture. We also added AI-assisted capabilities where they provide operational value, while keeping deterministic and auditable fallbacks so the system remains useful even when an external AI provider is unavailable.

Agenda

• Project architecture and the problem being solved.

• Your contribution: HRMS and Recruitment.

• Your contribution to AI features across HRMS, Recruitment, Expenses, and Inventory.

• Your friend's contribution: Expenses and Inventory Management.

• Integrated demonstration flow, validation, and future plan.

Page 2 - Architecture, Security, and Common Platform

Speech: The common platform is built around a secure organization context. Every authenticated request is associated with an organization, and database queries are scoped by organization ID. This prevents one tenant from seeing another tenant's employees, candidates, expenses, inventory, or AI audit records.

The frontend is built with React 19, TypeScript, Vite, Tailwind CSS, Framer Motion, and TanStack Query. TanStack Query helps us manage loading states, caching, invalidation, and refetching after mutations. The backend uses Express controllers, services, middleware, and Mongoose models. This separation keeps UI behavior, business logic, and persistence responsibilities clear.

Layer	Main responsibility
Frontend	Responsive dashboards, forms, drawers, cards, tables, charts, and AI result views.
API and controllers	Authentication, validation, request handling, response contracts, and module routes.
Services	Business rules such as leave accruals, payroll calculations, recruitment scoring, forecasting, and OCR.
MongoDB	Tenant-scoped operational records, immutable inventory ledger data, and AI audit runs.
RBAC and audit	Module-level permissions, action permissions, approval workflows, and traceability.

A key design principle is graceful degradation. When Gemini or Groq is unavailable, the system does not invent an answer or silently fail. It uses a verified deterministic result where appropriate and reports provider status to the user.

Page 3 - HRMS Module: Workforce Foundation

Speaker: You

Speech: I worked on the HRMS module, which manages the employee lifecycle after a person joins the organization. HRMS begins with departments, designations, branches, employees, reporting relationships, and employee documents. It provides a 360-degree employee view rather than storing only a name and email address.

The employee dossier brings together job information, department, manager, branch, skills, statutory details, bank and payroll details, education, previous work history, attendance, leave, and documents. This makes HRMS useful to HR teams, managers, finance teams, and employees themselves.

The module also supports an interactive organization chart. Managers can understand reporting relationships, direct reports, total team size, and branch distribution. This is useful for workforce planning and for quickly navigating from an organizational node to an employee dossier.

Core HRMS features

- Employee master data and dual identity linking between login users and personnel profiles.
- Departments, designations, pay grades, branches, managers, skills, and career history.
- Employee documents and a structured 360-degree dossier.
- Interactive organization chart and reporting hierarchy.
- Role-based access for HR, managers, employees, and administrators.

The important contribution here was connecting detailed HR records with real operational workflows instead of treating HRMS as a static employee directory.

Page 4 - HRMS Module: Attendance, Leave, Payroll, and Performance

Speech: HRMS includes an attendance engine with multiple shifts, grace periods, weekly-off rules, live check-in and check-out status, attendance regularization, and compensatory-off handling. If an employee misses a punch, the employee can submit a regularization request for manager approval rather than editing attendance directly.

The leave engine is policy-driven. It supports leave balances, monthly accruals, carry-forward limits, sandwich leave detection, encashment, and an audit ledger. Every credit and debit is traceable, which is important because leave balances affect employee trust and payroll accuracy.

The payroll engine uses salary structures, earnings, statutory deductions, attendance-driven loss-of-pay calculations, approval stages, payslip generation, and employee self-service access. The goal is to reduce manual spreadsheet calculations and connect payroll with real attendance and approved leave data.

The performance area supports goals and OKRs with weighted key results, milestones, progress updates, check-ins, review cycles, and appraisals. This helps connect employee activity with measurable business outcomes.

Area	What to explain during the demo
Attendance	Live punch, shift status, regularization request, approval.
Leave	Balance, policy calculation, approval, accrual ledger.
Payroll	Salary structure, LOP calculation, approval, PDF payslip.
Performance	Goal, key result, milestone, progress, review cycle.

Page 5 - Recruitment Module: Complete Applicant Lifecycle

Speech: I also worked on the Recruitment or Applicant Tracking System module. Recruitment covers the entire talent lifecycle: manpower requisition, job opening, publication, candidate profile, application pipeline, interviews, scorecards, offer management, and onboarding.

A recruiter can create a job opening with role details, skills, experience, location, employment type, salary range, and hiring approvals. Candidates can be maintained as reusable profiles and connected to one or more job applications. The pipeline provides a Kanban-style view so recruiters can move candidates through defined stages while preserving activity history.

Interviews are structured rather than informal. Interview panels, schedules, question kits, competency ratings, notes, recommendations, and feedback are captured in the system. This reduces subjective decision-making and makes the final hiring decision explainable.

Job offers include compensation, joining details, approval status, offer documents, and acceptance tracking. After acceptance, the ATS-to-HRMS bridge can convert a candidate into an employee and carry forward relevant details and documents. This removes duplicate data entry during onboarding.

Recruitment features to highlight

- Requisitions and multi-step approval workflows.
- Job openings, candidate profiles, applications, and pipeline stages.
- Interviews with competency scorecards and question kits.
- Job offers, approval, acceptance, and joining workflow.
- Talent pools, candidate assignment, and job-opening selection.
- One-click bridge from accepted candidate to HRMS employee.

Page 6 - AI Features: Architecture and Responsible Use

Speaker: You

Speech: In addition to HRMS and Recruitment, I worked on the AI feature analysis, integration, and testing across all modules. The AI design uses a provider gateway where external providers are used only where they add value. Recruitment primarily uses Groq and Gemini through a controlled gateway. Expenses and selected Inventory functions use Gemini. Several features are intentionally deterministic because calculations such as totals, thresholds, leave balances, and inventory valuation must be verifiable.

The system validates inputs and outputs, records AI runs with organization, user, task, model, latency, and result summary, and applies tenant scoping. If a provider fails or reaches quota, the user receives a verified fallback or an explicit status message. This is safer than presenting a fabricated AI answer.

Module	AI and intelligence approach
Recruitment	Groq/Gemini generation, parsing, match scoring, question kits, marketing, and copilot; deterministic fallback.
Expenses	Gemini-assisted receipt and natural-language parsing plus rules for policy, budget, duplicate, and anomaly chec
Inventory	Forecasting, OCR, anomaly detection, supplier scoring, and read-only copilot using ledger-backed data.
HRMS	Deterministic goal/performance intelligence and HR copilot-style workforce summaries.

AI should assist a responsible user, not replace approval, audit, or human review. This principle is visible throughout the project.

Page 7 - Recruitment AI Features in Detail

Speech: Recruitment contains the largest set of generative AI workflows. First, the system can generate a job description and recruitment marketing content from structured job information. It can create LinkedIn-style posts and social variants in different tones while preserving the actual role requirements.

Second, resume parsing extracts structured candidate information such as contact details, experience, skills, education, and a summary. This reduces manual data entry, but the recruiter still reviews the extracted result before saving it.

Third, match scoring compares a candidate's profile with the job requirements. The result includes an overall score, strengths, gaps, and a recommendation. This supports consistent shortlisting, but it is not an automatic hiring decision.

Fourth, interview question-kit generation creates questions based on the role and competencies. The interview panel can then record ratings and notes in a standardized scorecard.

Finally, Recruitment Copilot can answer recruiter questions and synthesize context from jobs, candidates, applications, and interviews. Talent pool support helps recruiters group candidates for future opportunities.

Recruitment AI feature list

- AI job description generation.
- LinkedIn and social recruitment post generation.
- Resume and candidate information parsing.
- Candidate-to-job match scoring with strengths and gaps.
- Role-specific interview question-kit generation.
- Recruiter Copilot synthesis and talent-pool assistance.
- Provider gateway, validation, fallback, and AI audit logging.

Page 8 - Expenses Module: Friend's Contribution

Speaker: Friend

Speech: I worked on the Expenses module. Its purpose is to control the complete expense lifecycle from category and budget setup to claim submission, approval, reimbursement, payment tracking, and reporting.

Expense categories and department budgets provide financial structure. Employees or authorized users can submit claims with amount, date, category, department, client or project context, billable status, receipts, and notes. Approval workflows ensure that claims are reviewed before reimbursement.

The module supports policy checks such as duplicate claims, out-of-policy amounts, missing receipts, budget utilization, and approval authority. Reimbursement and payment states make it possible to distinguish submitted, approved, rejected, reimbursed, and paid records.

The system also provides dashboards and summaries so finance teams can view spending by department, category, period, and status. This makes expenses a controlled business process rather than a collection of uploaded bills.

Expense features

- Expense categories and department budget configuration.
- Claim submission with receipts and structured metadata.
- Multi-stage approval and rejection with audit history.
- Policy, duplicate, anomaly, and budget checks.
- Reimbursement, payment tracking, and activity history.
- Department, category, time-period, and executive summaries.

Page 9 - Expenses AI and Intelligence

Speech: The Expenses module combines generative extraction with deterministic controls. Receipt scanning can use Gemini to extract merchant, date, amount, tax, currency, category, line items, and confidence. The structured result is shown for user review before it becomes an expense claim.

Natural-language expense parsing allows a user to type a sentence such as, 'Dinner with Acme Corp for 85 dollars yesterday.' The service extracts the amount, date, category, billable status, client, and confidence. Structured JSON output is requested so the backend can validate and normalize the result.

Other intelligence features are rules-based by design. Duplicate detection compares key claim signals. Policy checks compare claims against configured limits. Budget alerts detect departments approaching or exceeding their budgets. Anomaly detection highlights unusual amounts, repeated merchants, or suspicious patterns. These results are more explainable when implemented through explicit rules.

AI safety point: receipt and natural-language extraction are suggestions, not final accounting entries. A user must review the output, and the claim remains subject to approval and audit.

Suggested demo

- Open an expense claim and show structured fields.
- Run receipt scan and point out confidence and manual review.
- Enter a natural-language expense sentence.
- Show the parsed result and budget/policy validation.
- Open Spend Copilot and ask for a department spending summary.

Page 10 - Inventory Management: Friend's Contribution

Speaker: Friend

Speech: I worked on the Inventory Management module. It covers items and categories, suppliers, warehouses, zones and bins, stock movements, batches, procurement, goods receipt, returns, assets, and analytics.

The most important technical design is the append-only StockLedger. Instead of overwriting stock history, the system records receipts, issues, transfers, adjustments, and reversals as events. Current stock and valuation can therefore be explained from transactions. This improves auditability and helps prevent overselling or silent stock corrections.

The procurement workflow connects purchase requisitions, purchase orders, goods receipt notes, supplier invoices, and three-way matching. The system can compare ordered quantity, received quantity, and invoiced quantity, while tolerance rules identify discrepancies.

Inventory also supports FIFO and weighted-average valuation, batch and serial tracking, warehouse traceability, equipment kits for HRMS onboarding, supplier scorecards, stock aging, turnover, and executive analytics.

Inventory features

- Item catalog, categories, suppliers, warehouses, zones, and bins.
- Append-only stock ledger and traceable stock movements.
- FIFO and weighted-average valuation.
- Purchase requisition, purchase order, GRN, returns, and three-way match.
- Batch, expiry, serial, asset, and equipment-kit workflows.
- ABC-XYZ analysis, aging, turnover, DSI, supplier metrics, and exports.

Page 11 - Inventory AI Features in Detail

Speech: Inventory AI begins with demand forecasting. The service analyzes historical issue transactions, calculates a run rate, estimates days of supply, identifies urgency, and recommends a reorder quantity. The user can generate a draft purchase requisition from the forecast, but it remains a draft and follows the normal approval process.

The OCR workflow accepts an invoice or challan image, extracts document details and line items, and can compare the result against a purchase order. Unmatched items, quantity differences, and price variance are surfaced as discrepancies for review.

Anomaly detection scans for shrinkage spikes, unusual adjustments, duplicate receipts, dead stock, stockout risks, and other operational signals. Users can acknowledge, dismiss, resolve, or mark a false positive, creating a human-in-the-loop feedback trail.

Supplier scoring combines delivery, acceptance quality, price stability, and consistency. The score is useful for procurement decisions and supplier review meetings.

Inventory Copilot is read-only. It answers questions about low stock, valuation, warehouses, expiring batches, and supplier spend. Its values are calculated from tenant-scoped records and reconciled with the append-only ledger. If Gemini is unavailable or quota-limited, the verified rule-based answer remains available and the user is informed.

Page 12 - Recommended Live Demonstration Sequence

Speech: We will now demonstrate one connected business journey rather than opening random screens.

Step	Action	Point to explain
1	Open dashboard and organization context	Common platform, role permissions, tenant-scoped data.
2	Open Recruitment job and candidate pipeline	Stages, activities, candidate context, and interview workflow.
3	Run Recruitment AI match or question kit	AI assistance, review, confidence, and auditability.
4	Accept an offer and show HRMS bridge	Candidate-to-employee conversion and reduced duplicate entry.
5	Open HRMS employee dossier	Attendance, leave, documents, goals, and reporting line.
6	Submit or scan an expense	Extraction is a draft; policy and approval remain authoritative.
7	Open Inventory Copilot	Ledger-backed valuation, low stock, warehouse, and fallback status.
8	Run forecast and draft requisition	AI recommendation feeds a controlled procurement workflow.

Validation statement

We tested the Inventory AI suite end to end with 11 of 11 tests passing, including forecasting, OCR, anomaly feedback, Copilot intents, supplier scoring, audit logging, and tenant isolation. Recruitment AI suites and frontend production builds were also validated. These tests do not replace user acceptance testing, but they demonstrate that the core paths are connected and repeatable.

Page 13 - Future Plan for the Whole Project

Speech: Our future plan is to evolve Nebula from a strong academic ERP prototype into a production-ready business platform.

Priority	Future enhancement	Expected value
1	AI observability dashboard with provider, model, latency, confidence, fallback, and token usage	Users and administrators can understand every AI result.
2	Provider governance with quotas, retries, circuit breakers, model failover, and cost controls	More reliable AI with predictable operating cost.
3	Workflow automation and notifications for approvals, expiring batches, leave, and low-inventory alerts	Less manual follow-up and faster decisions.
4	Advanced analytics and forecasting across HR, recruitment, expenses, and inventory	Cross-module planning and better executive insight.
5	Stronger testing: unit, integration, end-to-end, accessibility, load, and security tests in CI/CD	Safer releases and easier maintenance.
6	Mobile and offline-friendly employee and warehouse experiences.	Better usability for field and operational teams.
7	Accounting, payroll banking, tax, email, calendar, and document-management integrations	Closer alignment with real enterprise operations.
8	Secret management, key rotation, encryption, retention policies, and production-grade security	Production readiness and governance.

We also plan to improve data quality controls so that operational summaries consistently reconcile with source ledgers, as demonstrated in the Inventory Copilot work.

Page 14 - Closing and Questions

Speaker: You

Speech: To conclude, Nebula HUB combines HRMS, Recruitment, Expenses, Inventory, CRM, and analytics in a common multi-tenant platform. My contribution focused on HRMS, Recruitment, and the AI capabilities across the project. My friend focused on Expenses and Inventory Management. Together, we connected operational workflows with approvals, auditability, analytics, and responsible AI assistance.

The central lesson from this project is that enterprise software is not only about screens. It is about reliable data, secure access, traceable workflows, clear ownership, and useful decisions. AI becomes valuable when it is integrated into those workflows and when the user can still review and trust the result.

Final line: Nebula HUB is our foundation for a connected, intelligent, and accountable business operations platform. Thank you. We are happy to answer your questions.

Quick question preparation

• Why did you use deterministic fallbacks? - To preserve verified operation when providers fail and prevent fabricated results.

• How is tenant isolation enforced? - Organization context is attached to authenticated requests and included in database filters and AI records.

• Is AI making final decisions? - No. AI recommends or extracts; users and approval workflows remain responsible.

• What was the most important integration? - Recruitment to HRMS onboarding, and Inventory ledger data feeding trustworthy AI summaries.

• What is the next major step? - Production hardening: observability, security, CI/CD, integrations, and cross-module analytics.